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Hybrid Multiscale Simulation Enhanced by Super – Resolution for Multiphase Flow in High – Contrast Heterogeneous Reservoirs

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Abstract

Robust field development strategies rely on optimizing well placement and control through extensive high-resolution reservoir simulations. However, the computational cost of such simulations can be prohibitively high. To address this challenge, this study develops an efficient dynamic well-control optimization workflow by leveraging a super-resolution deep neural network (SR-DNN) model to alleviate the computational burden. The SR-DNN is trained on paired coarse- and fine-scale reservoir simulations to learn the nonlinear mapping between coarse-scale state variables (pressure and saturation) and their fine-scale counterparts. The SR-DNN is integrated with an efficient coarse-scale simulator, forming a hybrid forward model that accurately predicts production well rates. The workflow is validated through dynamic well-control optimization for a water-flooding process in a channelized reservoir. The SR-DNN achieves excellent prediction accuracy, with average root mean square error (RMSE) of 76.1 psi for pressure and 0.046 for saturation throughout 10 years of prediction horizon. The structural similarity index measure (SSIM) further confirms high fidelity, averaging 0.9899 for pressure and 0.9129 for saturation. The hybrid forward model demonstrates strong agreement with fine-scale simulations, yielding L_2 -norm relative error of 1.79% and 5.54% for oil and water production cumulative, respectively. Additionally, the hybrid model achieves an NPV with an R^2 score of 0.92 relative to the fine-scale simulation benchmark, while delivering a $16\times$ speedup with an upscaling factor of 5×5 . This proposed work features a novel integration of deep-learning-based super-resolution with coarse-scale reservoir simulation which can be readily used for fine-scale dynamic well-control optimization. Unlike purely data-driven approaches, the method preserves physical consistency by leveraging coarse-scale simulations for guidance. The proposed framework advances reservoir simulation by combining computational efficiency with high accuracy, enabling large-scale well-control optimization with practical feasibility.

Introduction

Numerical reservoir simulators have served as indispensable tools in modern reservoir management, supporting daily reservoir engineering tasks such as production forecast, history matching, and production optimization. These simulators were implemented to solve the complex multiphase flow equations in porous

media (Lie, 2019), with accuracy fundamentally depends on the spatial resolution of the computational grid. To capture subsurface heterogeneities and localized flow dynamics (e.g., viscous fingering effect), reservoir simulation engineers typically employ high-resolution models with millions to billions of grid cells. However, such fine-scale simulations incur prohibitive computational costs in CPU time and memory resources, especially when they are repeatedly evaluated in iterative workflows like history matching and production optimization. Traditionally, high-performance computing techniques have advanced simulation speed through parallelized linear solvers (Killough & Wheeler, 1987) or nonlinear solvers (Cao et al., 2005), the implementation of these sophisticated solvers usually requires multi-discipline domain expertise, and such simulators have been required to run on high-performance hardware like CPU or GPU-based supercomputers.

Additionally, coarse-scale modeling employs upscaling techniques with different numerical homogenization procedures to address computational challenges, while preserving geological features (Durlofsky & Chen, 2012; King et al., 2006). The accuracy of these methods essentially depends on the spatial domain (local versus global) to compute the coarse properties (e.g., transmissibility), and whether single-phase or multiphase flow functions are incorporated. Besides, upscaling methods have been customized in different aspects, such as boundary conditions, upscaling calculations, computational domain size, and the specific coarse-scale equations. The computation cost rising from upscaling can be compensated by the improved accuracy and computational efficiency. However, even complicated flow-based upscaling approaches (Siavashi et al., 2022) cannot fully eliminate errors stemming from lost fine-scale details, which can impact simulation accuracy and potentially bias decisions for reservoir management.

Recent advances in deep learning have offered new possibilities to directly accelerate large-scale reservoir simulations while maintaining satisfactory physical fidelity. Zhu and Zabaras (2018) and Wang et al. (2020) demonstrated the success of convolutional neural networks (CNNs) in subsurface flow modeling, and further Tang et al. (2020) and Yan et al. (2022) incorporated temporal dynamics and physics-based constraints to deal with 2D and 3D reservoir simulations. Physics-informed neural networks (Raissi et al., 2019) further enhanced the integration of governing equations into data-driven neural networks, improving generalization and physical consistency, while the training cost of such models can be typically higher than pure data-driven training methods. The ultimately trained deep learning models can be directly used as surrogate forward models for inverse modeling tasks like history matching (Aslam et al., 2025), or production optimization (Han et al., 2025).

Further, machine and deep learning methods have been increasingly employed to enhance the accuracy of coarse-scale models while maintaining computational efficiency compared to fine-scale simulations. These approaches typically focus on correcting errors introduced by coarse-scale approximations. Trehan & Durlofsky (2018) developed a random forest-based post-processing framework to estimate upscaling errors across multiple geological realizations, significantly improving oil production forecasts and uncertainty quantification. Tang & Durlofsky (2022) further advanced this by training tree-based models to correct low-fidelity (LF) model errors, enabling efficient well-placement optimization while preserving accuracy. However, such methods often lack fine-scale predictions of reservoir state variables. More recently, deep generative models have been applied to bridge this gap. Liu et al. (2023) introduced Simulation Enhancement GAN (SE-GAN), which reconstructs fine-scale dynamics from coarse-scale outputs, achieving notable gains in computational speed and accuracy. Li et al. (2025) developed a super-resolution GAN for multi-scale reservoir simulation, successfully predicting key state variables (e.g., pressure, saturation). Similarly, Kim et al. (2025) leveraged a diffusion model for coarse-model error correction. Despite these advances in fine-scale simulation, current methods have not yet tackled with both reservoir and well-based state variables at fine scale, which can be critical requirement for production optimization and high-resolution simulations.

This study addresses these limitations by developing a super-resolution deep learning framework specifically designed for fine-scale reservoir simulation. Our approach leverages coarse-scale simulation

results to predict corresponding fine-scale pressure and saturation fields, effectively constructing a hybrid surrogate model that combines computational efficiency with high-fidelity accuracy.

Methodology

Simulation Model Description

The physical problem considered in this study is a two-phase water flooding process in a two-dimensional (2D) heterogeneous reservoir. The reservoir domain is in rectangular shape. Two injection wells are drilled nearby one edge of the reservoir, while two additional production wells are drilled nearby the edge in the opposite direction. No-flowing boundary conditions are imposed on the four outer edges of the reservoir. The reservoir domain is initially saturated with oil and water phases, with oil being the primary phase. The two-phase fluid flow in the reservoir follows Darcy's law and the mass conservation equation for each fluid phase (o for oil, w for water). The governing equations, which account for fluid storage, fluid flux, and source/sink terms, can be written as,

$$\frac{\partial (\phi \rho_\alpha S_\alpha)}{\partial t} - \nabla \cdot \left(\frac{\rho_\alpha K k_{ra}}{\mu_\alpha} (\nabla p_\alpha - \rho_\alpha g \nabla z) \right) - \rho_\alpha q_\alpha = 0 \quad (1)$$

where ϕ and K are rock porosity and intrinsic permeability, respectively; k_{ra} , ρ_α and μ_α are relative permeability, density and viscosity for phase α , respectively; S_α and p_α represent saturation and pressure for phase α , respectively; t denotes time; g is the gravitational acceleration; z represents depth; and q_α is the fluid source/sink term.

The fluid phases in the porous media satisfy the saturation constraint, shown as below,

$$S_w + S_o = 1 \quad (2)$$

In this study, the effect of capillary pressure between oil and water is ignored, such that,

$$p_c = p_w - p_o = 0 \quad (3)$$

The above equations are solved through finite difference method with typical reservoir simulators, and ultimately obtain pressure and saturation of each phase at each time step and grid blocks. In this work, the commercial reservoir simulator CMG IMEX ([Computer Modelling Group Ltd, 2024](#)) is adopted to perform the simulation at both coarse and fine-scale discretization, in order to prepare training and testing data for the SR-DNN model.

Super-Resolution Deep Neural Network Model

In this section, a full-scale super-resolution deep neural network (SR-DNN) model is introduced to predict reservoir dynamics at the fine scale based on coarse-scale simulation outputs. The architecture of SR-DNN is a multi-stage configuration designed to handle $5\times$ upscaling ($20 \times 20 \rightarrow 100 \times 100$) while preserving both local details and global flow patterns essential for accurate reservoir simulation. As shown in [Fig. 1](#), it adopts a unified architecture that processes complete coarse-scale simulation outputs to reconstruct full fine-scale reservoir fields. The network consists of four main components: feature extraction at the coarse and fine scales, residual enhancement, spatial transformer blocks, and upscaling with fine-grid integration.

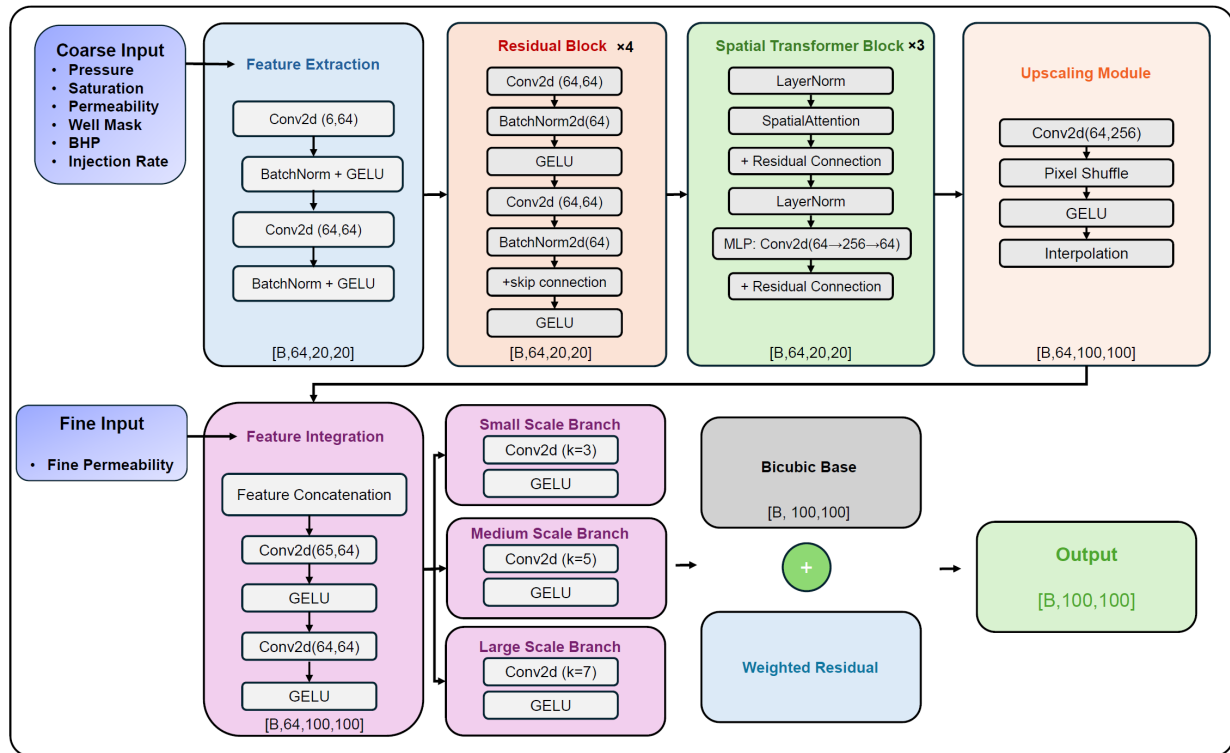


Figure 1—The architecture of the super-resolution deep neural network (SR-DNN) model

Feature Extraction: the feature extraction module processes the coarse-scale input tensor, including the target field (pressure or saturation), water saturation, permeability, well location mask, bottom-hole pressure mask, and injection rate mask. This module employs a dual-layer convolutional architecture that maps the 6-channel input tensor to 64 feature channels through 3×3 convolutions with GELU activation, ensuring comprehensive feature capture while maintaining spatial relationships.

Residual Block: the residual enhancement network comprises four consecutive residual blocks, each of which contains dual convolutional layers followed by batch normalization and skip connections. This configuration enables hierarchical feature learning while effectively preventing gradient degradation through stable gradient flow paths.

Spatial Transformer Block: three spatial transformer blocks are integrated to capture global dependencies and long-range correlations essential for reservoir flow physics. Each spatial transformer block implements multi-head self-attention mechanisms (4 heads) adapted for 2D spatial data, enabling the network to understand flow connectivity and well interference patterns across the entire simulation domain.

Upscaling and Fine-Grid Integration: The upscaling module employs a hierarchical approach for $5\times$ scaling. The network applies a single PixelShuffle operation ($2\times$) that increases resolution from 20×20 to 40×40 , followed by bicubic interpolation ($2.5\times$) to reach the final 100×100 resolution, ensuring optimal feature preservation throughout the upscaling process.

A novel aspect of our architecture is the integration of fine-scale permeability information. After upscaling, the network concatenates the upscaled features with fine-scale permeability data, processing this combined information through a specialized integration module that enhances reconstruction accuracy by incorporating high-resolution geological constraints.

Output Generation: The final reconstruction employs a multi-scale residual learning strategy that predicts corrections to a bi-cubic interpolation baseline. Three parallel branches with different receptive fields (2D convolution kernel sizes of 3×3 , 5×5 , 7×7) capture small-scale details, medium-scale features, and large-scale patterns respectively. The final prediction combines these multi-scale residuals with a learnable weighting parameter:

$$y_{\text{fine}} = I_{\text{base}} + \alpha \cdot (r_{\text{small}} + r_{\text{medium}} + r_{\text{large}}) \quad (4)$$

where I_{base} represents bi-cubic interpolation of the coarse target field; r_s denote multi-scale residual predictions at each scale s ; α is a learnable parameter.

Loss Function: the training strategy employs a comprehensive multi-component loss function designed to ensure both global accuracy and preservation of critical flow features:

$$L_{\text{total}} = L_{\text{MSE}} + \lambda_{\text{well}} L_{\text{well}} + \lambda_{\text{grad}} L_{\text{grad}} + \lambda_{\text{TV}} L_{\text{TV}} + \lambda_{\text{diff}} L_{\text{diff}} \quad (5)$$

The loss function consist of four sub-terms: (1) L_{MSE} denotes the standard mean square error loss between prediction and ground truth for overall accuracy; (2) L_{well} is the enhanced well location loss ($\lambda_{\text{well}} = 1000$) that emphasizes accuracy at critical well positions; (3) L_{grad} is the gradient preservation loss ($\lambda_{\text{grad}} = 2$) to maintain sharp boundaries and flow patterns; (4) L_{TV} is the total variation regularization ($\lambda_{\text{TV}} = 0.1$) for smoothness; and (5) L_{diff} is the difference enhancement loss ($\lambda_{\text{diff}} = 0.5$) that improves learning of residual corrections relative to the bi-cubic baseline.

This comprehensive architecture achieves computational efficiency with approximately 2.8×10^6 trainable parameters while enabling accurate reconstruction of complete fine-scale reservoir simulation fields. The framework is implemented in PyTorch with mixed-precision training and gradient scaling for optimal performance.

Net Present Value Calculation

As part of evaluating the surrogate model accuracy for techno-economic analysis, we calculate the Net Present Value (NPV) for each case. NPV serves as a representative objective function commonly used in applications such as dynamic well-control optimization.

$$NPV = \sum_{n=1}^{N_t} \frac{1}{(1+b)^{n-1}} \left[\sum_{i=1}^{N_{\text{prod}}} (c_o \cdot q_{oi}^n - c_w \cdot q_{wi}^n) - \sum_{i=1}^{N_{\text{inj}}} (c_{iw} \cdot q_{iw,i}^n) \right] \quad (6)$$

where N_t is the number of time steps of NPV evaluation (in years); N_{prod} and N_{inj} are the number of production and injection wells, respectively; q_{oi}^n , q_{wi}^n , $q_{iw,i}^n$ are the oil production rates, water production rates, and water injection rates of well i at time step n , respectively; c_o is the oil selling price; c_w is the processing cost of produced water; c_{iw} is the operational cost of injected water; b is the annual discount rate.

Numerical Experiments

Reservoir Model Description

In this study, we employed two-dimensional channelized reservoir models with domain size of $10,000 \times 10,000 \times 10 \text{ ft}^3$. This physical system forms the basis of the super-resolution problem, where we aim to reconstruct fine-scale (100×100) pressure and saturation fields from coarse-scale (20×20) simulation results. The reservoir model is initially saturated with oil and water with uniform oil saturation of 0.9 and a reservoir pressure of 4800 psia. All sides of the rectangular domain are subject to no-flow (closed) boundary conditions. Fig. 2 presents three pairs of sample permeability realizations at both coarse and fine scales, out of 500 generated sample pairs. The general setup consists of two channel bodies located on the left and right sides of the reservoir, each intersected by an injection–production well pair (Inj1–Prd1 and Inj2–Prd2). Well placements are fixed across all cases and are kept consistent between the fine and coarse models.

A bimodal permeability distribution is used to capture the contrast between sand and mud facies, with mean permeabilities of 897 mD (sand) and 30 mD (mud), respectively. This significant contrast directs the injected water flow toward the high-permeability sandy channels, which predominantly control water-flooding performance. This poses a challenge for coarse-scale models, as the upscaling process tends to

smear out the channel geometries, leading to notable discrepancies in model predictions between the coarse and fine resolutions.

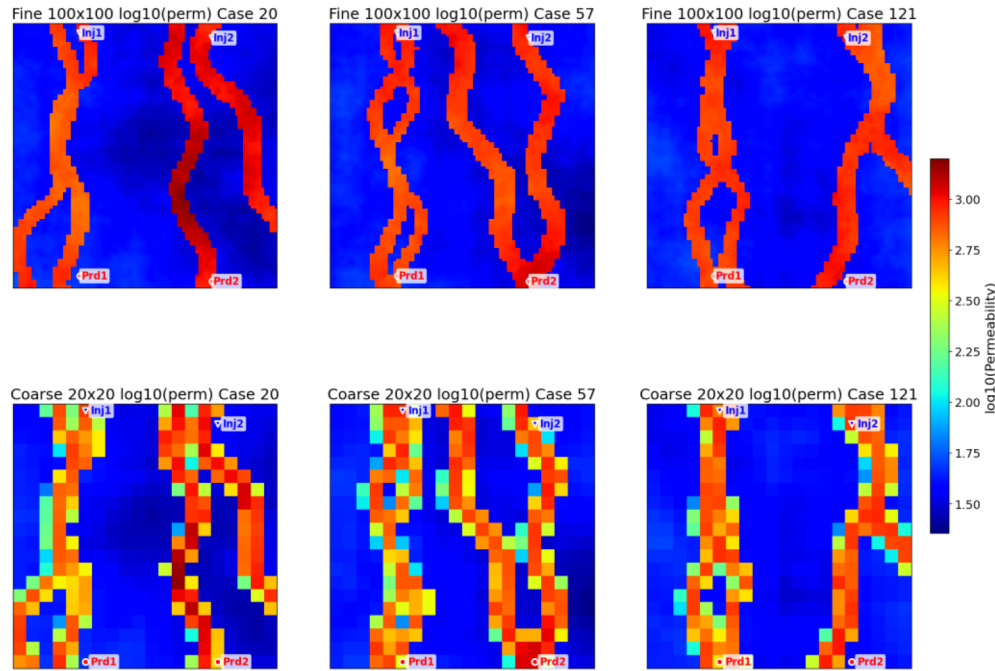


Figure 2—Three example cases of log permeability fields of the bimodal channelized reservoir model sample realizations. Top row: fine scale model (100 × 100 grid blocks), coarse scale model (20 × 20 grid blocks).

For all simulation samples, we used a constant porosity, $\phi = 0.15$. The fluid system is modeled as immiscible two-phase (oil–water) flow, with a constant water viscosity of 0.31 *cp*. Oil viscosity, is pressure-dependent, with a reference value of 0.91 *cp* at 515 psi. The relative permeability curves follow the Brooks–Corey model, using an exponent of 2 for both oil and water phases, and are illustrated in Fig. 3. Each simulation spans a total duration of 3650 days, with outputs recorded at 365-day intervals. Table 1 provide a summary of the reservoir model parameters and simulation setup used to generate the dataset.

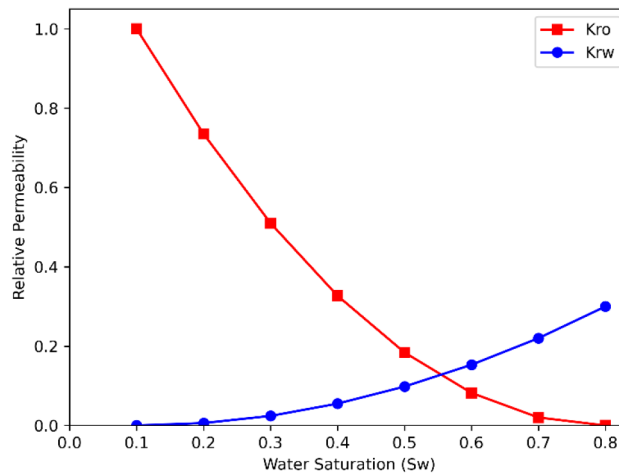


Figure 3—Relative permeability curve employed in the coarse-fine scale simulation.

Table 1—Reservoir model parameters.

Parameters	Values	Units
Reservoir domain size	$10,000 \times 10,000 \times 10$	ft^3
Fine-scale grid block number	$100 \times 100 \times 1$	-
Coarse-scale grid block number	$20 \times 20 \times 1$	-
Rock compressibility, c_r	3×10^{-6}	$psia^{-1}$
Water density, ρ_w	62.2	lb_m/ft^3
Water viscosity, μ_w	0.31	cP
Oil density, ρ_o	46.2	lb_m/ft^3
Oil viscosity, μ_o	0.91	cP
Initial oil saturation, S_o^{init}	0.9	Fraction
Initial reservoir pressure, p^{init}	4800	$psia$
Total reservoir life	3650	$Days$
Time step size, Δt	365	$Days$
Water injection rate	500 – 1000	STB/Day
Bottom-hole pressure in oil production wells	1500 – 2500	$psia$

Throughout the simulations, the injection well operates under a time-varying injection rate constraint, while the production wells are controlled using time-varying bottom-hole pressures. To increase the variance of reservoir model responses, we randomly sample both injection and production constraints at every time step using Latin Hypercube Sampling (LHS). The injection rate is bounded between 500 and 1000 *STB/day*, and the bottom-hole pressures for the production wells range from 1500 to 2500 *psia*. Fig. 4 shows an example of the time-varying injection and production control settings used in one of the simulation cases.

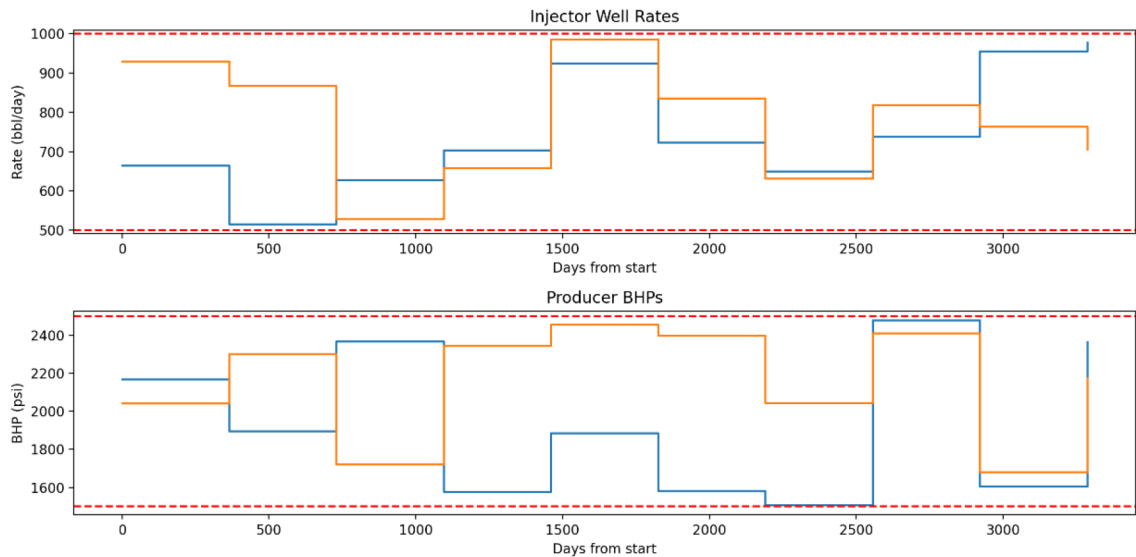


Figure 4—Example of time-varying injection and production control from one simulation case. Red-dashed line show the sampling bounds for each constraints.

Performance of SR-DNN at the Reservoir Scale

The Root Mean Squared Error (RMSE) and Structural Similarity Index Matrix (SSIM) are used to evaluate the accuracy of SR-DNN model at predicting the fine-scale pressure and saturation as compared to the ground truth from fine-scale simulation by reservoir simulator. RMSE provides a measure of the

average magnitude of prediction error, while SSIM quantifies the structural similarity between predicted fields and ground-truth fields, capturing both local patterns and global spatial consistency. These metrics were computed at each time step throughout the simulation period to assess both temporal and spatial performance.

All comparison are consistently done through all 50 testing cases that were not used in the training of SR-DNN. As a result, the forward model becomes hybrid, which combines SR-DNN with coarse-scale simulation in order to perform fine-scale simulation, which hereafter denoted "Coarse + SR-DNN" for simplicity. The counterpart of coarse- and fine-scale simulations are just denoted as "Coarse" and "Fine" simulation, respectively. Fig. 5 presents the mean RMSE and SSIM distribution of Coarse + SR-DNN from all test samples over the entire simulation period.

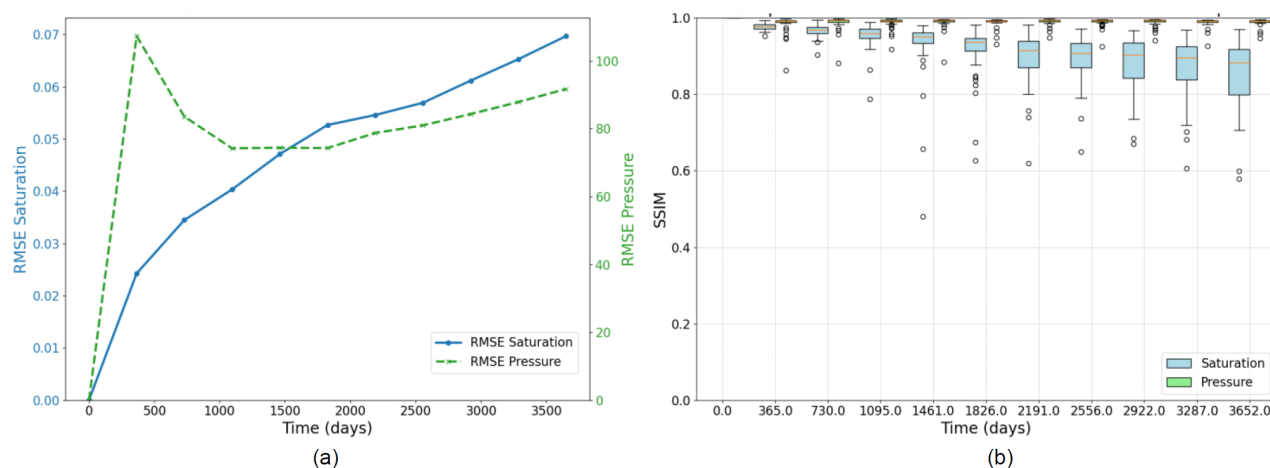


Figure 5—The temporal change of RMSE (a) and SSIM (b) for the prediction by Coarse + SR-DNN.

Overall, the Coarse + SR-DNN shows decent accuracy with average RMSE of 76.1 psi for pressure and 0.046 for saturation throughout 10 years of prediction horizon. The SSIM also confirms good spatial consistency of the prediction, with average SSIM of 0.9899 for pressure and 0.9129 for saturation. From the accuracy metrics trends throughout the simulation, we can see that the pressure prediction accuracy is generally stable with time, while saturation prediction shows decreasing accuracy with time. This can be attributed to the higher sensitivity of saturation to permeability heterogeneity, particularly in channelized systems. The presence of sharp saturation fronts and nonlinear flow dynamics makes it more challenging for the Coarse + SR-DNN model to maintain stable accuracy over time, compared to pressure which tends to evolve more rapid and continuously.

Fig. 6 presents the spatial (2D) comparison of pressure and saturation predictions among the Coarse simulation, Fine simulation, and Coarse + SR-DNN. Relative error maps are also included to assess the accuracy of Coarse + SR-DNN predictions against the Fine simulation reference. The results show that the Coarse + SR-DNN produces consistent predictions with the fine simulation, particularly in capturing the overall pressure distribution, which is consistent with Fig. 5(b). Saturation predictions are also reasonably accurate. In particular, the Coarse + SR-DNN is able to improve the localization of the sharp displacement front, which is typically blurred in the coarse simulation due to numerical dispersion. This improvement is largely attributed to the inclusion of both fine- and coarse-scale permeability maps as input features, allowing the model to better understand channel geometry and flow connectivity. However, the largest saturation errors still occur near the advancing water saturation front, indicating the difficulty in capturing sharp, nonlinear flow behavior. Despite this, the overall saturation distribution comparison shows that the Coarse + SR-DNN achieves significantly better reconstruction of fine-scale water-flood sweeping pattern,

which highlight its suitability as a reliable and computationally efficient surrogate for fine-scale predictions of reservoir state variables.

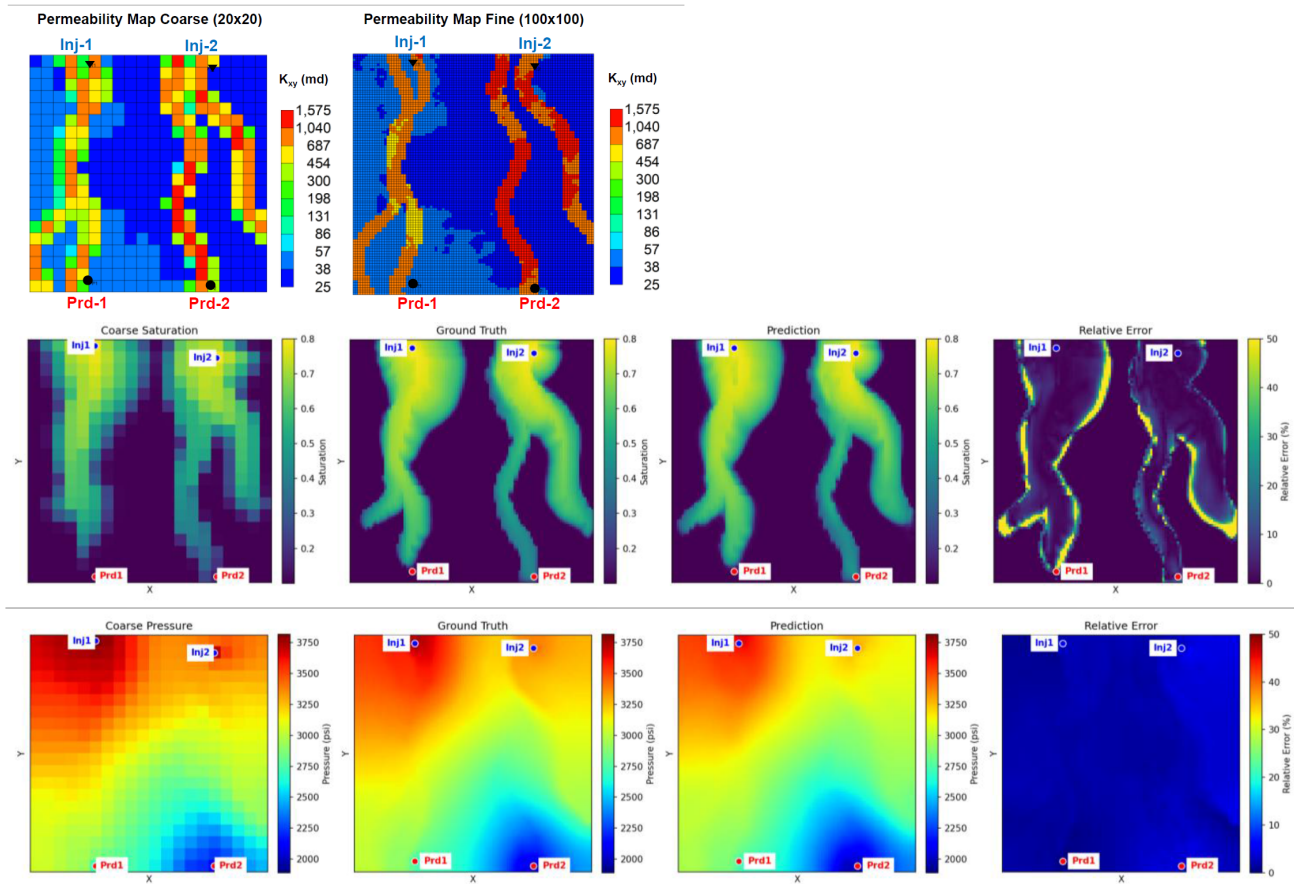


Figure 6—Pressure and saturation prediction comparison at coarse and fine scales among Coarse simulation (1st column), fine simulation (2nd column), Coarse + SR-DNN (3rd column), and relative error of Coarse + SR-DNN (4th column). The corresponding coarse and fine-scale permeability fields are presented in the 1st row, the saturation comparison is shown in 2nd row, and the pressure comparison is in 3rd row.

Performance of SR-DNN for Production Forecast

To assess the efficacy of the Coarse + SR-DNN as a hybrid surrogate model, we compare its predictions of the cumulative oil and water production profiles with those obtained from Fine simulations. Instead of using pressure and saturation at the well grid blocks, cumulative production is estimated using a material balance approach. This method reduces sensitivity to localized grid effects and provides a more consistent basis for evaluation. Production is calculated from the changes in in-place fluid volumes and the injected water volume, derived from the full-field pressure and saturation distributions, rather than relying on individual well production outputs such as those computed using the Peaceman well model (Peaceman, 1978). It is important to note that this approach is valid only for bounded reservoirs. Yet it can totally fit the purpose to calculate the final objective like Net Present Value, which is only related to field level injection and production flow rates.

Fig. 7 presents the box plots of relative error distributions, computed using the L_2 norm, for cumulative oil and water production from both the Coarse simulation and the Coarse + SR-DNN predictions, compared against the Fine simulation results. The Coarse + SR-DNN model shows a significant improvement in accuracy over the coarse simulation. Specifically, the relative errors for cumulative oil and water production using SR-DNN are $1.79 \pm 0.88\%$ and $5.54 \pm 3.05\%$, respectively, while the Coarse simulation only achieved $27.88 \pm 7.36\%$ and $84.66 \pm 36.25\%$ for the oil and water production. The much higher error in cumulative

water production observed in the coarse model signifies the impact of numerical dispersion, which tends to smear the water front and leads to inaccurate prediction of water breakthrough timing and sweep efficiency. The improved accuracy of cumulative production from the Coarse + SR-DNN is benefited from its superior capability to reconstruct fine-scale pressure and saturation fields.

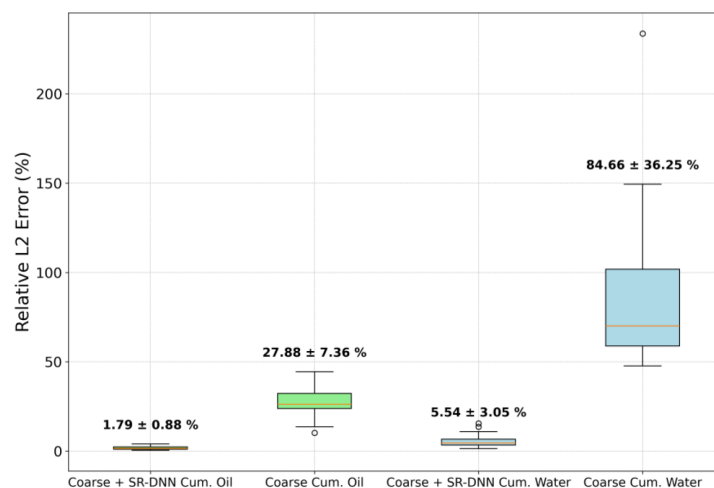


Figure 7—Boxplot of relative errors in cumulative oil and water production for the test dataset, comparing predictions from the Coarse + SR-DNN model and the Coarse simulation model only. Relative errors are computed using the L_2 norm with respect to the Fine simulation results.

Fig. 8 shows the comparison of cumulative oil and water production over time for the Coarse simulation, the SR-DNN, and the Fine simulation. The Coarse + SR-DNN remain consistent with the Fine simulation results throughout the entire simulation period. Notably, the Coarse + SR-DNN accurately captures the timing and behavior of water breakthrough observed in the Fine simulation model (X intercept of the blue curves, which in turn leads to more accurate predictions of cumulative oil production compared to the coarse simulation after the production wells start experiencing two-phase production. This capability is particularly important for techno-economic evaluations such as Net Present Value (NPV), where accurate rate–time profiles are more critical than simply matching the ultimate recovery.

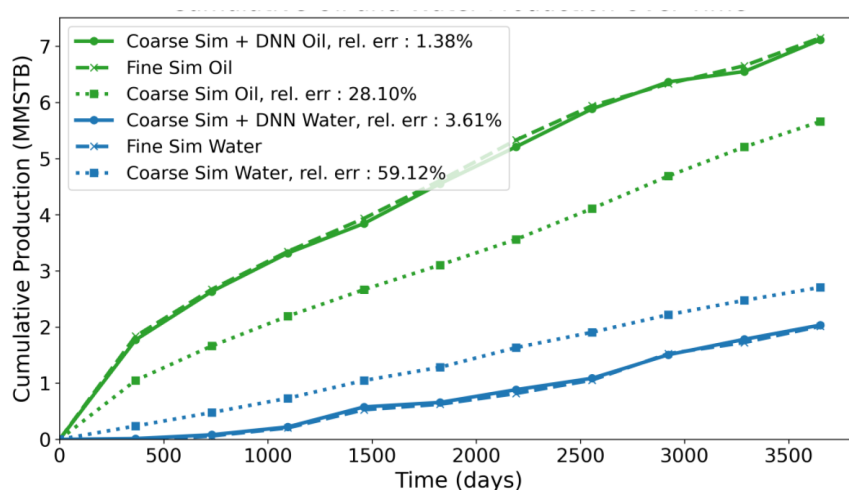


Figure 8—Cumulative production prediction comparison between Coarse + SR-DNN (solid lines), Coarse simulation (dotted square lines), and Fine simulation (dashed cross lines) based on an exemplary testing Case 20.

As the ultimate goal is to perform dynamic well-control optimization, the prediction accuracy of economic metrics like Net Present Value can be extremely important during the online iterative optimization

phase. Therefore, the NPV of all the 50 testing cases are calculated with the cumulative production streams from different predictive models based on Eq. (6). In this study, we assume an oil price of 70 \$/bbl, injection water cost of 3 \$/bbl, produced water handling cost of 5\$/bbl and an annual discount rate of 10%.

Fig. 9 presents a parity plot comparing the NPV values from the Coarse simulation and the Coarse + SR-DNN predictions against the Fine simulation results. The Coarse simulation (blue circles) generally under-predicts NPV due to premature water breakthrough, which can drop oil revenue yet increase water processing cost. In contrast, the Coarse + SR-DNN achieves much more accurate NPV predictions, with an R^2 score of 0.92 relative to the fine-scale benchmark. These results support the Coarse + SR-DNN as a hybrid surrogate model is a reliable reservoir simulation approach for forecasting and optimization.

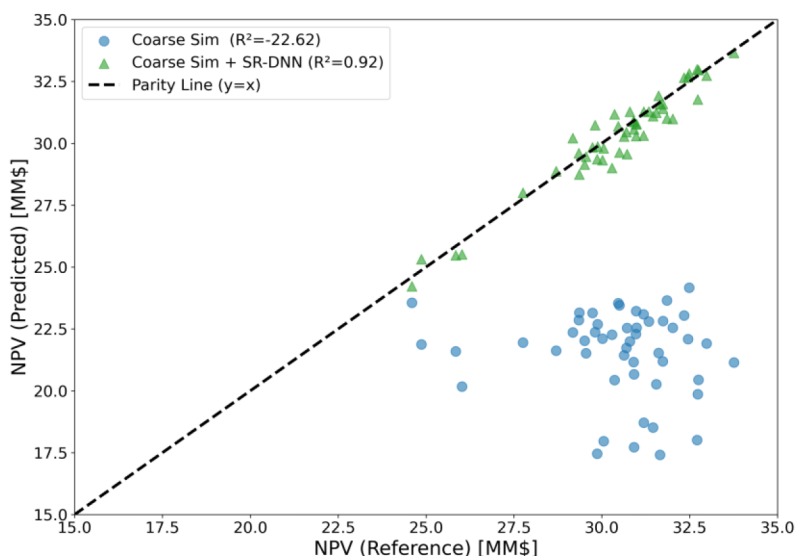


Figure 9—Parity plot of NPV comparison based on all the testing cases. X: NPV from Fine simulation (reference); Y: NPV from Coarse simulation, and Coarse + SR-DNN.

On the other hand, as production optimization is an iterative process with massive evaluations of forward reservoir models, the computational efficiency for such models makes a big difference in optimization efficiency. To examine the prediction efficiency of different forward models, we compare the CPU times of the Fine simulation, Coarse simulation, and Coarse + SR-DNN simulation across the entire test dataset, as shown in the box plots in Fig. 10. The mean CPU times are 11.08 seconds for the Fine simulation, 0.69 seconds for the coarse simulation, and 0.68 seconds for the combined Coarse + SR-DNN model.

It is worth noting that the mean prediction time for the SR-DNN alone (excluding the coarse simulation) is only 0.0072 seconds when evaluated in a batch mode on a CUDA-enabled GPU environment (NVIDIA RTX A4500), making its contribution to the total runtime practically negligible. Therefore, the overall speedup (approximately 16× in this 2D case) is driven primarily by the difference between coarse and Fine simulation runtimes. As a result, this hybrid-surrogate approach is to achieve improved prediction accuracy while significantly reducing computational cost in prediction. To further justify the deployment of super-resolution-based surrogates like SR-DNN, the number of fine simulations used for training should ideally be much smaller than the number of fine-scale evaluations required in iterative workflows such as optimization. We consider this should be a less concern, especially when dealing with stochastic optimization.

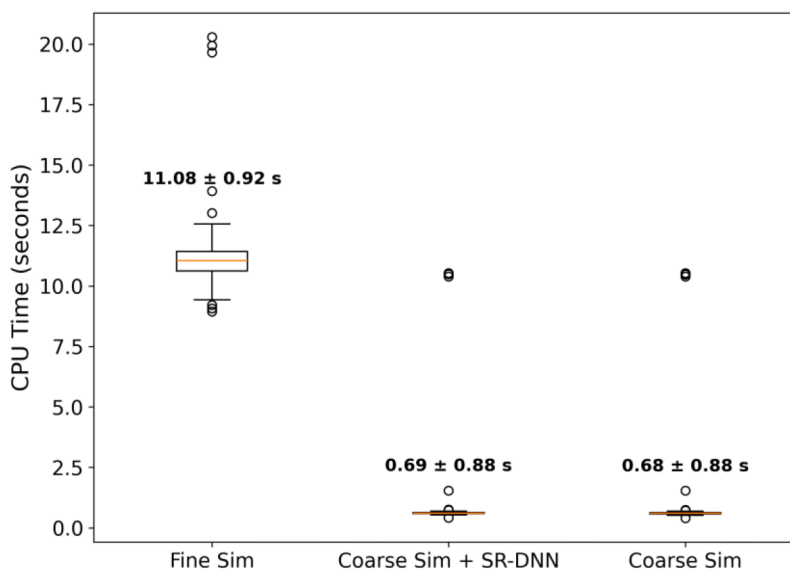


Figure 10—Comparison of prediction CPU time between Fine scale simulation, Coarse + SR-DNN, and Coarse simulation. Notice that the mean CPU time for SR-DNN inference is 0.0072 seconds.

Conclusion

In this study, a hybrid deep-learning-based surrogate modeling framework is developed through combining coarse-scale reservoir simulation with a super-resolution deep neural network (SR-DNN). Specifically, SR-DNN reconstructs fine-scale pressure and saturation fields based on coarse-scale simulation outputs, and ultimately calculation production profiles and high-level Net Present Value. The proposed approach addresses the accuracy bottleneck of conventional coarse-scale simulation and efficiency constrain of fine-scale simulation.

The SR-DNN is experimented through a relatively complex bi-modal channelized reservoir model with fixed well locations but time-varying well control. It is trained through paired coarse- and fine-scale simulation reservoir dynamics. The SR-DNN model demonstrates strong predictive performance across 50 test cases, achieving high structural similarity (SSIM) of 0.9899 and 0.9129 for pressure and saturation fields, respectively. Spatial comparisons further confirm that SR-DNN effectively corrects coarse model inaccuracies, particularly in capturing displacement water front location and overall sweep efficiency. These improvements are especially important in channelized systems, where the high – contrast heterogeneity plays a critical role in fluid front propagation behavior.

In addition to spatial accuracy, the combination of coarse simulation and SR-DNN model (Coarse + SR-DNN) demonstrates strong performance in forecasting production behavior. Cumulative oil and water production estimates derived using a material balance approach show close agreement with fine-scale simulation results. Compared to the coarse simulation, the SR-DNN reduces the relative errors in cumulative oil and water production from $27.88 \pm 7.36\%$ and $84.66 \pm 36.25\%$ to $1.79 \pm 0.88\%$ and $5.54 \pm 3.05\%$, respectively. Furthermore, the model provides reliable economic predictions, achieving an R^2 of 0.92 in Net Present Value (NPV) compared to the fine-scale benchmark, confirming its reliability for techno-economic assessments. In the current 2D model setup with 5×5 upscaling factor, Coarse + SR-DNN can achieve around 16 times speedup as compared to the Fine simulation models, and it is expected to bring more computational gain when dealing with larger reservoir model size and 3D reservoir models through even higher upscaling factor. Even though the current study focuses on forward modeling, the demonstrated accuracy and efficiency of the Coarse + SR-DNN indicates the confidence of its readiness for dynamic well-control optimization, through integrating with existing optimization workflows.

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Reference

- Aslam, B., Zhang, Y., Hoteit, I. and Yan, B. (2025). A Deep Neural Network–Based History-Matching Method for Deterministic Estimation of Heterogeneous Model Parameters. *SPE Journal*, pp.1-21.
- Cao, H., Tchelepi, H.A., Wallis, J. and Yardumian, H. (2005). Parallel scalable unstructured CPR-type linear solver for reservoir simulation. SPE Annual Technical Conference and Exhibition.
- Computer Modelling Group Ltd. (2024). *IMEX (Version 2024.20) [Computer software]*.
- Das, S., & Suganthan, P. N. (2011). Differential Evolution: A Survey of the State-of-the-Art. *IEEE Transactions on Evolutionary Computation*, **15**(1), 4–31. <https://doi.org/10.1109/TEVC.2010.2059031>
- Durlofsky, L. J. and Chen, Y. (2012). Uncertainty Quantification for Subsurface Flow Problems Using Coarse-Scale Models. In *Numerical Analysis of Multiscale Problems, Lecture Notes in Computational Science and Engineering*, 163–202. Springer.
- Han, Z., Tariq, Z. and Yan, B. (2025). A Novel Robust Optimization Framework Based on Surrogate Modeling for Underground Hydrogen Storage in Depleted Natural Gas Reservoirs. *InterPore Journal (Accepted)*.
- Killough, J. E., and M. F. Wheeler. (1987). Parallel iterative linear equation solvers: An investigation of domain decomposition algorithms for reservoir simulation. SPE Reservoir Simulation Conference.
- Kim, Y.D., Suzuki, S., Wu, X.H. and Becker, G. (2025). Diffusion Model-Based Multi-Fidelity Reservoir Simulation Surrogates for Upscaled Model Error Correction. In SPE Reservoir Simulation Conference.
- King, M.J., Burn, K.S., Wang, P., Muralidharan, V., Alvarado, F., Ma, X. and Datta-Gupta, A. (2006). Optimal coarsening of 3D reservoir models for flow simulation. *SPE Reservoir Evaluation & Engineering*, **9**(04), pp.317-334.
- Li, H., Aslam, B. and Yan, B. (2025). Multiscale Reservoir Simulation Through Super-Resolution Techniques. In SPE Gas & Oil Technology Showcase and Conference.
- Lie, Knut-Andreas. (2019). *An introduction to reservoir simulation using MATLAB/GNU Octave: User guide for the MATLAB Reservoir Simulation Toolbox (MRST)*. Cambridge University Press.
- Liu, Y., Yang, S., Zhang, N., Cao, J. and Guo, C., 2024. Simulation Enhancement GAN for Efficient Reservoir Simulation at Fine Scales. *Mathematical Geosciences*, **56**(7), pp.1439-1468.
- Peaceman, D. W. (1978). Interpretation of well-block pressures in numerical reservoir simulation (includes associated paper 6988). *Society of Petroleum Engineers Journal*, **18**(03), 183-194.
- Price, K. V., Storn, R. M., & Lampinen, J. A. (2005). *Differential evolution: A practical approach to global optimization*. Springer.
- Raissi, M., Perdikaris, P., & Karniadakis, G. E. (2019). Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, **378**, 686-707.
- Siavashi, J., Najafi, A., Ebadi, M., & Sharifi, M. (2022). A CNN-based approach for upscaling multiphase flow in digital sandstones. *Fuel*, **308**, 122047.
- Tang, H., & Durlofsky, L.J. (2022). Use of low-fidelity models with machine-learning error correction for well placement optimization. *Computational Geosciences*, **26**, 1189–1206. <https://doi.org/10.1007/s10596-022-10153-7>
- Tang, M., Liu, Y., & Durlofsky, L. J. (2020). A deep-learning-based surrogate model for data assimilation in dynamic subsurface flow problems. *Journal of Computational Physics*, **413**, 109456.
- Trehan, S., & Durlofsky, L.J. (2018). Machine-learning-based modeling of coarse-scale error, with application to uncertainty quantification. *Computational Geosciences*, **22**, 1093–1113. <https://doi.org/10.1007/s10596-018-9740-x>
- Wang, N., Zhang, D., Chang, H., & Li, H. (2020). Deep learning of subsurface flow via theory-guided neural network. *Journal of Hydrology*, **584**, 124700.
- Yan, B., Harp, D. R., Chen, B., Hoteit, H., & Pawar, R. J. (2022). A gradient-based deep neural network model for simulating multiphase flow in porous media. *Journal of Computational Physics*, **463**, 111277.
- Zhu, Y., & Zabaras, N. (2018). Bayesian deep convolutional encoder–decoder networks for surrogate modeling and uncertainty quantification. *Journal of Computational Physics*, **366**, 415-447.
- Zou, A., Ye, T., Volkov, O., & Durlofsky, L. J. (2022). Effective treatment of geometric constraints in derivative-free well placement optimization. *Journal of Petroleum Science and Engineering*, **215**, 110635. <https://doi.org/10.1016/j.petrol.2022.110635>